13. Conclusion

"We're entering a new world in which
data may be more important than software."
— Tim O'Reilly

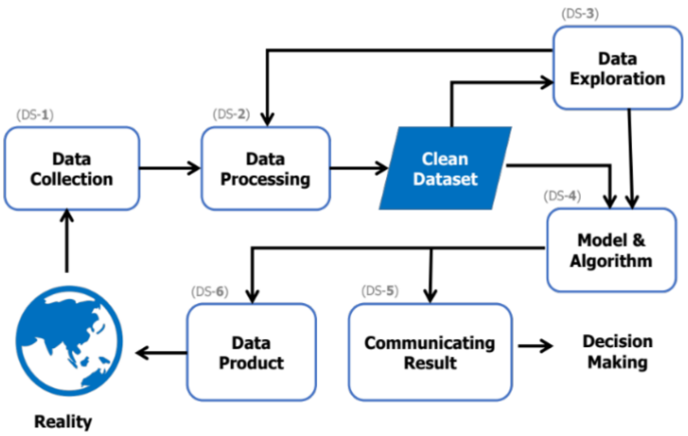

Figure 13.0-1: The data analytics process in the data science process

This book has explained data analytics through the lens of the data science process, following the framework in Doing Data Science: Straight Talk from the Frontline (O'Neil & Schutt, 2013), shown in Figure 13.0-1 — the structure guiding this book's chapters, each covering one stage, paired with Python code so readers with a programming background can put data analytics into real practice.

Reality

Reality is the ongoing activity of everything — living and non-living, on Earth and beyond, past, present, and future — constantly generating enormous volumes of data every fraction of a second. This stage was covered in Chapter 2, Data Collection, leading into the data collection process.

Data Collection (DS-1)

Covered in Chapter 2, Data Collection: how data is obtained from reality through recording, especially via Internet of Things (IoT) technology, which has made data collection far more efficient. This chapter also covered structured vs. unstructured data, data collection practices, the datasets used throughout this book, and importantly the four data scales — Nominal (Classification), Ordinal, Interval, and Ratio — so readers can understand data characteristics well enough to frame problems and transform data appropriately for their chosen technique.

Data Processing (DS-2)

Covered in Chapter 3, Data Processing: since real-world analytics work rarely gets ready-to-use data, this chapter covered cleaning and transforming data into a usable form, plus Python programming with NumPy and Pandas — a foundation for the Python-based analytics code throughout the rest of the book.

Clean Dataset

A clean dataset is data ready for analysis — the output of the data processing stage covered in Chapter 3.

Data Exploration (DS-3)

Covered in Chapter 4, Data Exploration: reviewing basic statistical functions — central tendency and dispersion — and using charts to explore data distribution and relationships between variables. This stage helps readers understand data behavior and plan an effective analytics strategy going forward.

Model Development (DS-4)

Covered across Chapters 5, 6, 7, 8, and 9. Chapter 5, Analytics Approach, defined what a model means, how to frame a received problem to guide technique and evaluation-method choices, and the process for building an effective, trustworthy model. Once the analytics direction is set, Chapter 6, Regression Analysis, Chapter

7, Classification Analysis, Chapter 8, Cluster Analysis, and Chapter 9, Recommender Systems, each covered core concepts, model-building techniques, evaluation methods, and Python code examples. Chapter 11, Additional Techniques, further covered techniques for specific data challenges and approaches to unstructured data analysis.

Communicating Results (DS-5)

Covered in Chapter 10, Data Visualization: helping readers understand the principles of presenting analytics results through charts and visuals in place of text, leveraging human visual perception, and covering how to turn data into images so audiences can quickly grasp key points and act on them decisively.

Data Product (DS-6)

Building a data product means turning a model into an actual deployed product. Chapter 9, Recommender Systems, laid out a path for building recommender systems that readers can extend into their own product features. The research featured in Chapter 12, Research Case Studies, showed how models from analytics were implemented as software components within larger systems or projects.

To connect the full picture — from upstream data, through midstream analysis into a model, to downstream deployed results — Chapter 12, Research Case Studies, presented 4 of the author's own research projects: Detecting Road Damage with a Gyro Sensor, Detecting Kicks with a Microwave Sensor, A Recommender for Biologists Studying Fungi–Host Pairs, and A Recommender for Vehicle Rest Stops — each applying data-analytics science to solve a real problem.

Since data analytics keeps advancing and evolving, readers can follow updates — datasets, code examples, full-color figures, and content revisions for this book — at:

`https://github.com/Rathachai/DA-LAB`